F005 | First experimental-data review

v0.2.0 uploaded results / v0.3.0 integrity and baseline patch | 19 October 2026

Evidence from the supplied F005 pilot

The robot sloshing arrays were reported as matching 9,822 paired trajectories with 1,500 samples each; dt = 0.004 s, acquisition window = 5.996 s. The small archive MD5 passed. The 49.9 GB archive MD5 remains uncomputed.

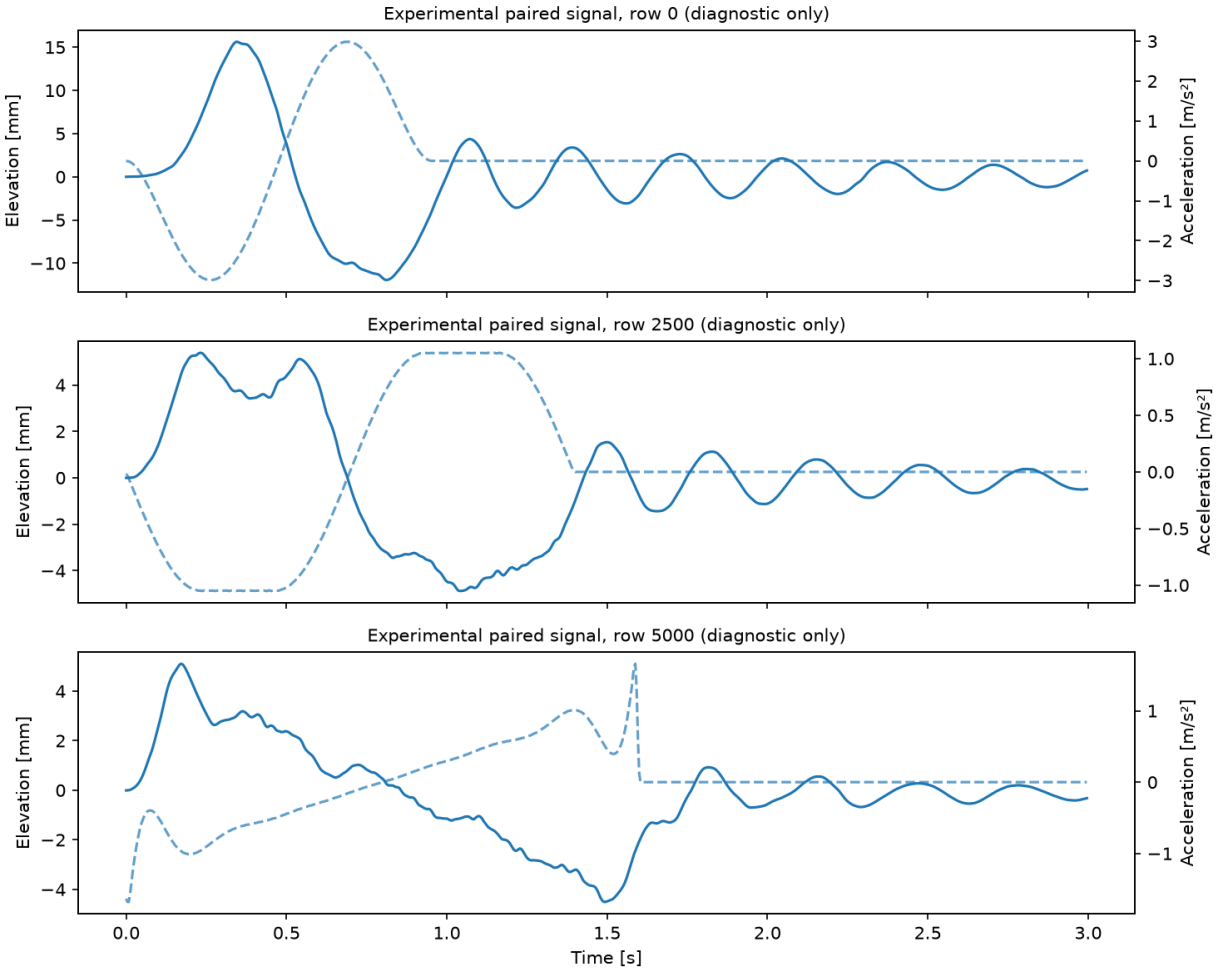
Audit item	Observed result	Status
Split row IDs	0..9821 exactly once	PASS structure
Candidate counts	3,500 / 750 / 750 / 4,822	PASS structure
Robot selected sample	131 rows, all finite	PASS preview
Approx. FFT-bin spacing	0.333333 Hz (750 samples)	Diagnostic only
NoBody pressure member	CSV CRC verified; metadata missing	UNVERIFIED units
Group/leakage audit	No trajectory metadata uploaded	NOT RUN
Euler / PKLSA predictions	No independent comparison yet	NOT RUN

Scientific interpretation

After input acceleration becomes approximately zero, the measured wall elevation oscillates while its amplitude decreases. This is consistent with forced sloshing followed by damped free response. It is NOT a demonstrated Kelvin wave, bulk inertial wave, or independent SST effect.

The prior pilot FFT maximum was evaluated only on the first 750 samples (3 seconds). The corresponding 1/3 Hz frequency-bin separation is too coarse to report a certified eigenfrequency; spectral peaks also depend on input forcing and windows.

Sample observations (not model predictions)



Blue: measured water elevation (mm); dashed: imposed acceleration (m/s2), on separate y-axis. The plot depicts three selected trajectories, not a population-wide estimate.

Next-run science contract

P5 checks exact partition coverage, repeated metadata IDs/labels, duplicate input arrays across partitions, and heuristic similar-shape collisions. Without source-defined trajectory families, P5 cannot be certified blind/disjoint.

P6 development only: fit a linear ARX(2) acceleration-response surrogate on the 3,500 candidate training trajectories; assess one-step error on 750 candidate validation trajectories. Do not access test/OOD target values. ARX is not a full Euler solver.

Limits: no PKLSA inference, no gravitational claims; original source ZIPs unchanged. This report is a review of uploaded pilot outputs only; original NPY arrays are available only on the user workstation.